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User equipment (UE) status information delivery in a wireless communication network

Abstract

A data communication system authenticates a wireless communication device, and in response, determines status information for the wireless communication device. The data communication system receives a telephony request for the wireless communication device. The data communication system delivers a telephony service to the wireless communication device in response to the telephony request and based on the status information.

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Background/Summary

RELATED CASES (1) This United States Patent Application is a continuation of U.S. patent application Ser. No. 17/493,181 that was filed on Oct. 4, 2021 and is entitled “USER EQUIPMENT (UE) STATUS INFORMATION DELIVERY IN A WIRELESS COMMUNICATION NETWORK.” U.S. patent application Ser. No. 17/493,181 is hereby incorporated by reference into this United States Patent Application.

TECHNICAL BACKGROUND
(1) Wireless communication networks provide wireless data services to wireless user devices. Exemplary wireless data services include machine-control, internet-access, media-streaming, and social-networking. Exemplary wireless user devices comprise phones, computers, vehicles, robots, and sensors. The wireless user devices execute user applications that use the wireless data services. For example, a smartphone may execute a social-networking application that communicates with a content server over a wireless communication network.
(2) The wireless communication networks have wireless access nodes which exchange wireless signals with the wireless user devices over radio frequency bands. The wireless signals use wireless network protocols like Fifth Generation New Radio (5GNR), Long Term Evolution (LTE), Institute

of Electrical and Electronic Engineers (IEEE) 802.11 (WIFI), and Low-Power Wide Area Network (LP-WAN). The wireless access nodes exchange network signaling and user data with network elements that are often clustered together into wireless network cores. The wireless network elements comprise Unified Data Repositories (UDRs), Unified Data Management (UDM), Access and Mobility Management Functions (AMFs), Session Management Functions (SMFs), User Plane Functions (UPFs), and the like.

(3) An Internet Protocol Multimedia Subsystem (IMS) typically includes an IP-based telephony system for placing and receiving voice calls. To use the IMS, the wireless user devices obtain IP addresses from their wireless communication networks and register their IP addresses and corresponding telephone numbers with the IMS. To place a call, the calling wireless user device transfers a Session Initiation Protocol (SIP) message to the IMS with the telephone numbers of the calling and called wireless user devices. The IMS translates the telephone number of the called user device into its IP address and transfers a SIP message with the telephone numbers of the calling and called wireless user devices to that IP address. If the called wireless user device accepts the call, the called user device returns a SIP message to the IMS with the call acceptance. In response to the call acceptance, the IMS transfers SIP messaging to the calling and called wireless user devices to indicate the IP address of the other device. The called and calling wireless user devices then exchange voice data using the IP addresses. In some scenarios, each wireless communication network has its own IMS and the IMSs interact to handle voice calls in a distributed manner. When a wireless user device visits another wireless communication network and its IMS, the visited network and visited IMS exchange signaling with the home network and home IMS to serve the visiting UE with voice calling.

(4) Within the IMS, a Telephony Application Server (TAS), manages the various communication paths that serve the voice call. Before selecting an access path for a wireless user device, the TAS first requests current status information for the wireless user device from the wireless communication network. The status information indicates voice-calling capabilities, access technology, and the like. The wireless communication network gathers and transfers the requested user information to the TAS. The TAS then selects the voice-calling access network based on the current status data for the wireless user device.

(5) Network Exposure Functions (NEFs) expose network information to other network elements like the current location and access network for a wireless user device. Application Functions (AFs) interact with the NEFs over Application Programming Interfaces (APIs) to obtain the network information. The AFs may transfer the network information for a wireless user device to an Application Server (AS) that is operated by an entity that controls the wireless user device.

(6) Unfortunately, the wireless communication networks do not efficiently and effectively gather status information for the wireless user devices. Moreover, the wireless communication networks force the network elements to wait before receiving the requested status information which delays service delivery to the wireless user devices.

TECHNICAL OVERVIEW

(7) In some examples, a wireless communication device is authenticated, and in response, status information for the wireless communication device is determined. A telephony request is received for the wireless communication device. A telephony service is delivered to the wireless communication device in response to the telephony request and based on the status information.

(8) In some examples, a first network system of the wireless communication network authenticates a wireless communication device, and in response, collects information related to the wireless communication device. A second network system of the wireless communication network receives a telephony invite related to the wireless communication device, and in response, sends a request for the information to the first network system The first network system transfers the information to the second network system in response to the request from the second network system. The second network system receives the information from the first network system, and in response, selects a

Radio Access Technology (RAT) type and indicates the selected RAT type to the wireless communication device. The first network system delivers a telephony service of the wireless communication network to the wireless communication device over the selected RAT type.
(9) In some examples, a wireless network system authenticates a wireless communication device, and in response, determines status information for the wireless communication device. A Session Initiation Protocol (SIP) system receives a request for the wireless communication device, and in response, retrieves the status information for the wireless communication device from the wireless network system and selects a Radio Access Technology (RAT) type for the wireless communication device based on the status information. The wireless network system establishes a communication link for the wireless communication device over the RAT type selected by the SIP system.

Description

DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an exemplary wireless communication network that serves User Equipment (UEs) and delivers UE status information for the UEs upon request.
(2) FIG. **2** illustrates an exemplary operation of the wireless communication network to serve the UEs and deliver the UE status information for the UEs upon request.
(3) FIG. **3** illustrates an exemplary operation of the wireless communication network to serve the UEs and deliver the UE status information for the UEs upon request.
(4) FIG. **4** illustrates an exemplary Fifth Generation (5G) wireless communication network that serves a UE and delivers UE status information for the UE upon request.
(5) FIG. **5** illustrates an exemplary UE in the 5G wireless communication network.
(6) FIG. **6** illustrates an exemplary 5G new radio access node in the 5G wireless communication network.
(7) FIG. **7** illustrates an exemplary wireless fidelity access node in the 5G wireless communication network.
(8) FIG. **8** illustrates an exemplary data center in the 5G wireless communication network.
(9) FIG. **9** illustrates an exemplary operation of the 5G wireless communication network to serve the UE and deliver the UE status information for the UE upon request.
(10) FIG. **10** illustrates an exemplary operation of the 5G wireless communication network to serve the UE and deliver the UE status information for the UE upon request.
(11) FIG. **11** illustrates an exemplary operation 5G wireless communication network to serve the UE and deliver the UE status information for the UE upon request.
DETAILED DESCRIPTION
(12) FIG. **1** illustrates exemplary wireless communication network **100** that serves User Equipment (UEs) **101-103** and deliver UE status information for UEs **101-103** upon request. Wireless communication network **100** comprises UEs **101-103**, wireless access nodes **111-113**, network controller **121**, Unified Data Management (UDM) **122**, Unified Data Repository (UDR) **123**, network element **124**. UEs **101-103** comprise computers, phones, vehicles, sensors, robots, or some other data appliance with communication circuitry. Network element **124** comprises an Application Function (AF), Network Exposure Function (NEF), Home Subscriber System (HSS), Telephony Application Server (TAS), and/or some other network apparatus. Network controller **121** comprises an Access and Mobility Management Function (AMF), Mobility Management Entity (MME), and/or some other network control-plane. Wireless communication network **100** is simplified for clarity and typically includes far more UEs, access nodes, and network elements than shown.
(13) Various examples of network configuration and operation are described herein. In some examples, network controller **121** exchanges registration signaling with UEs **101-103** over wireless access nodes **111-113**. In response, network controller **121** exchanges authentication signaling for

UEs **101-103** with UDM **122**. UDM **122** exchanges the authentication signaling with network controller **121**, and in response, UDM **122** transfers UDM requests for UE status information for individual UEs **101-103** to network controller **121**. Network controller **121** exchanges network signaling with UEs **101-103** and determines the UE status information. The UE status information indicates: a time of last radio contact with the UE, a Radio Access Technology (RAT) type serving the UE, Voice over Packet (VoP) capability for the UE, Internet Protocol Multimedia Subsystem (IMS) capability for the UE, Session Initiation Protocol (SIP) capability for the UE, geographic location for the UE, successful authentication for the UE, and/or some other UE metadata. Network controller **121** receives the UDM requests, and in response to the UDM requests, network controller **121** transfers the UE status information for individual UEs **101-103** to UDM **122**. UDM **122** receives the UE status information from network controller **121** and transfers the UE status information to UDR **123**. UDR **123** receives the UE status information from UDM **123**. UDR **123** receives a status request for an individual one of UEs **101-103** from network element **124** and responsively transfers the UE status information for the individual one of UEs **101-103** to network element **124**. Advantageously, wireless communication network **100** efficiently and effectively gathers status information for UEs **101-103**. Moreover, wireless communication network **100** serves the status information to network element **124** with a minimal wait to speed-up service-delivery to UEs **101-103**.

(14) UEs **101-103** and wireless access nodes **111-113** communicate over wireless links that use wireless technologies like Fifth Generation New Radio (5GNR), Long Term Evolution (LTE), Institute of Electrical and Electronic Engineers (IEEE) 802.11 (WIFI), Low-Power Wide Area Network (LP-WAN), Bluetooth, and/or some other wireless communication protocols. The components of wireless communication network **100** communicate over network connections that comprise metallic wiring, glass fibers, radio channels, or some other communication media. The network connections use technologies like IEEE 802.3 (ETHERNET), Internet Protocol (IP), Time Division Multiplex (TDM), Data Over Cable System Interface Specification (DOCSIS), General Packet Radio Service Transfer Protocol (GTP), 5GNR, LTE, WIFI, LP-WAN, Bluetooth, virtual switching, inter-processor communication, bus interfaces, and/or some other data communication protocols. UEs **101-103** and wireless access nodes **111-113** comprise radios. UEs **101-103**, wireless access nodes **111-113**, network controller **121**, UDM **122**, UDR **123**, and network element **124** comprise microprocessors, software, memories, transceivers, bus circuitry, and the like. The microprocessors comprise Digital Signal Processors (DSP), Central Processing Units (CPU), Graphical Processing Units (GPU), Application-Specific Integrated Circuits (ASIC), and/or the like. The memories comprise Random Access Memory (RAM), flash circuitry, disk drives, and/or the like. The memories store software like operating systems, user applications, radio applications, and network functions. The microprocessors retrieve the software from the memories and execute the software to drive the operation of wireless communication network **100** as described herein.

(15) FIG. **2** illustrates an exemplary operation of wireless communication network **100** to serve UE **101** and deliver UE status information for UE **101** upon request. Network controller **121** exchanges registration signaling with UE **101** over wireless access node **111** and responsively exchanges authentication signaling for UEs **101-103** with UDM **122** (**201**). UDM **122** exchanges the authentication signaling with network controller **121** (**202**). In response, UDM **122** transfers a UDM request for UE status information for UE **101** to network controller **121** (**202**). Network controller **121** exchanges network signaling with UE **101** and determines the UE status information (**203**). Network controller **121** receives the UDM request, and in response to the UDM request, network controller **121** transfers the UE status information for UE **101** to UDM **122** (**203**). UDM **122** receives the UE status information from network controller **121** and transfers the UE status information to UDR **123** (**204**). UDR **123** receives the UE status information from UDM **123** (**205**). UDR **123** receives a status request for UE **101** from network element **124** and responsively transfers the UE status information for UE **101** to network element **124** (**205**).

(16) FIG. **3** illustrates an exemplary operation of wireless communication network **100** to serve UEs **102-103** and deliver UE status information for UEs **102-103** upon request. UE **102** wirelessly attaches to wireless access node **112**. UE **102** and network controller **121** exchange registration signaling over wireless access node **112**. In response, network controller **121** and UDM **122** exchange authentication signaling for UE **102**. UDM **122** transfers a UDM request for UE status information for UE **102** to network controller **121** in response to the authentication of UE **102**. UE **102** and network controller **121** exchange network signaling. Network controller **121** determines UE status information for UE **102**. Network controller **121** transfers the UE status information for UE **102** to UDM **122**. UDM **122** transfers the UE status information for UE **102** to UDR **123**. UDR **123** receives a status request for UE **102** from network element **124** and responsively transfers the UE status information for UE **102** to network element **124**. In this example, network element **124** selects an access network type for UE **102** based on the UE status information for UE **102**. For example, network element **124** may select a different wireless communication network for UE **102** to use for a voice call based on UE capabilities and network status.
(17) UE **103** wirelessly attaches to wireless access node **113**. UE **103** and network controller **121** exchange registration signaling over wireless access node **113**. In response, network controller **121** and UDM **122** exchange authentication signaling for UE **103**. UDM **122** transfers a UDM request for UE status information for UE **103** to network controller **121** in response to the authentication of UE **103**. UE **103** and network controller **121** exchange network signaling. Network controller **121** determines UE status information for UE **103**. Network controller **121** transfers the UE status information for UE **103** to UDM **122**. UDM **122** transfers the UE status information for UE **103** to UDR **123**. Network element **124** receives a status request for UE **103** from an external system and responsively transfers a status request for UE **103** to UDR **123**. UDR **123** receives the status request for UE **103** from network element **124** and responsively transfers the UE status information for UE **103** to network element **124**. Network element **124** transfers the UE status information for UE **103** to the external system in response to the status request. For example, network element **124** may comprise a Network Exposure Function (NEF) and Application Function (AF) that serve UE status data to an external Application Server (AS).
(18) FIG. **4** illustrates an exemplary Fifth Generation (5G) wireless communication network **400** that serves UE **401** and delivers UE status information for UE **401** upon request. 5G wireless communication network **400** comprises an example of wireless communication network **100**, although network **100** may differ. 5G wireless communication network **400** comprises: UE **401**, 5GNR Access Node (AN) **411**, WIFI AN **412**, and data center **420**. Data center **420** comprises Interworking Function (IWF) **421**, Access and Mobility Management Function (AMF) **422**, Session Management Function (SMF) **423**, Unified Data Management (UDM) **424**, Unified Data Repository (UDR) **425**, Network Exposure Function (NEF) **426**, User Plane Function (UPF) **427**, Authentication and Security Function (AUSF) **428**, Home Subscriber System (HSS) **429**, Call State Control Functions (CSCF) **430**, Telephony Application Server (TAS) **431**, and Application Function (AF) **432**. Wireless communication network **400** is simplified for clarity and typically includes far more UEs, ANs, data centers, and network functions than shown.
(19) UE **401** wirelessly attaches to 5GNR AN **411**. UE **401** and AMF **422** exchange registration signaling over 5GNR AN **411**. The registration signaling reports UE capabilities to AMF **422** like Voice over Packet (VoP) capability, Internet Protocol Multimedia Subsystem (IMS) capability, Session Initiation Protocol (SIP) capability, and geographic location. In response to the registration signaling, AMF **422** and UDM **424** exchange authentication signaling for UE **401** over AUSF **428**. In response to the authentication of UE **401**, UDM **424** subscribes to UE status information for UE **401** from AMF **422**. AMF **422** interacts with UDM **424** to select initial context for UE **401**. AMF **422** interacts with SMF **423** and other network functions to select slices, policies, addresses, and additional context for UE **401**. Part of the context characterizes an IMS bearer between UE **401** and CSCF **430** over 5GNR AN **411** and UPF **427**. SMF **423** directs UPF **427** to serve the IMS bearer

for UE **401**. AMF **422** directs 5GNR AN **411** to serve the IMS bearer for UE **401**. AMF **422** directs UE **401** to use the IMS bearer over 5GNR AN **411**. UE **401** and CSCF **430** exchange IMS registration signaling over the IMS bearer which traverses 5GNR AN **411** and UPF **427**. CSCF **430** and HSS **429** exchange IMS registration data for UE **401**.

(20) In response to the subscription for UE **401** from UDM **424**, AMF **422** identifies UE status information that indicates: a time of last radio contact, a RAT type (5GNR in this example), VoP capability, IMS capability, SIP capability, geographic location, successful authentication, and/or some other UE metadata. AMF **422** transfers the UE status data for UE **401** to UDM **424** to serve the subscription. UDM **424** transfers the UE status data for UE **401** to UDR **425**. UDR **425** may now deliver the UE status data for UE **401** to TAS **431**, NEF **426**, and/or some other network element upon request.

(21) UE **401** places a multimedia call by transferring a SIP invite for the called-party to CSCF **430** over the IMS bearer. CSCF **430** transfers the invite to TAS **431**. TAS **431** queries HSS **429** for the UE status information for UE **401**. HSS **429** retrieves the UE status information for UE **401** from UDR **425** and responds to TAS **431** with the UE status information for UE **401**. TAS **431** selects an access network type for UE **401** based on the UE status information for UE **401**. TAS **431** might select a Voice over New Radio (VoNR) network that traverses 5GNR AN **411** and UPF **427** based on a 5GNR VoP capability for UE **401**. TAS **431** might select Code Division Multiple Access (CDMA) fallback network **440** for UE **401** based on an inadequate VoP capability for UE **401**. TAS **431** informs CSCF **430** of the selected access network type for UE **401**. CSCF **430** informs AMF **422** of the selected access network type for UE **401**—typically over a Policy Control Function (PCF). AMF **422** signals UE **401** over 5GNR AN **411** to use the selected access network type. UE **401** typically uses the VoNR network that traverses 5GNR AN **411** and UPF **427** if UE **401** is VoNR capable and the VoNR network is available. UE **401** typically uses CDMA fallback network **440** when UE **401** is not VoNR capable or when the VoNR network is unavailable.

(22) UE **401** receives a multimedia call in a similar manner. CSCF **430** receives a SIP invite for UE **401** from another UE or CSCF. CSCF **430** transfers the invite to TAS **431**. TAS **431** obtains the UE status information for UE **401** as above. TAS **431** selects the access network type for UE **401** based on the UE status information as described above.

(23) In some examples, AS **450** has an affiliation with UE **401** like employer/employee, parent/child, sensor/owner, or the like. AS **450** logs-in to AF **432** and requests UE status information for UE **401**. AF **432** requests the UE status information for UE **401** from NEF **431**. NEF **431** requests the UE status information for UE **401** from UDR **425**. UDR **425** already has the UE status information for UE **401** as described above. UDR **425** transfers the UE status information for UE **401** to NEF **431**. NEF **431** transfers the UE status information for UE **401** to AF **432**. AF **432** transfers the UE status information for UE **401** to AS **432** for use by the employer, parent, or other affiliated entity.

(24) In some examples, UE **401** wirelessly attaches to WIFI AN **412**. UE **401** registers with IWF **421** over WIFI AN **412** and establishes a secure tunnel. IWF **421** and AMF **422** exchange registration signaling for UE **401**. AMF **422** and UDM **424** exchange authentication signaling for UE **401** over AUSF **428**. In response, UDM **424** subscribes to UE status information for UE **401** from AMF **422**. Aside from the use of WIFI AN **412** and IWF **421**, the operation for placing and receiving multimedia calls is like that described above—although the RAT type is now WIFI. The operation for AS **450** would also be similar. For example, AS **450** could use the UE status information to determine UE location and whether UE **401** is attached to 5GNR, WIFI, or some other RAT type at the location.

(25) FIG. **5** illustrates exemplary UE **401** in 5G wireless communication network **400**. UE **401** comprises an example of UEs **101-103**, although these UEs may differ. UE **401** comprises 5GNR radio **501**, WIFI radio **502**, CDMA radio **503**, user circuitry **504**, and user components **505**. User components **505** comprise sensors, controllers, displays, or some other user apparatus that

generates slice data. Radios **501-503** comprise antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, and transceivers that are coupled over bus circuitry. User circuitry **504** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in user circuitry **504** stores an operating system (OS), user applications (APP), and network applications for WIFI, 5GNR, and IP. The antennas in 5GNR radio **501** are wirelessly coupled to 5GNR AN **411** over a 5GNR link. The antennas in WIFI radio **502** are wirelessly coupled to WIFI AN **412** over a WIFI link. The antennas in CDMA radio **503** are wirelessly coupled to CDMA network **440** AN **412** over a CDMA link. Transceivers (XCVRs) in radios **501-503** are coupled to transceivers in user circuitry **504**. Transceivers in user circuitry **504** are coupled to user components **505**. The CPU in user circuitry **504** executes the operating system, user applications, and network applications to exchange network signaling and user data with ANs **411-412** and network **440** over radios **501-503**.

(26) FIG. **6** illustrates exemplary 5G New Radio (NR) Access Node (AN) **411** in 5G wireless communication network **400**. 5GNR AN **411** comprises an example of wireless access nodes **111-113**, although access nodes **111-113** may differ. 5GNR AN **411** comprises Radio Unit (RU) **601**, Distributed Unit (DU) **602**, and Centralized Unit (CU) **603**. RU **601** comprises 5GNR antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, radio applications, and transceivers that are coupled over bus circuitry. DU **602** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in DU **602** stores an operating system and 5GNR network applications (Physical Layer, Media Access Control, Radio Link Control). CU **603** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in CU **603** stores an operating system, IP, and 5GNR network applications (Packet Data Convergence Protocol, Service Data Adaptation Protocol, Radio Resource Control). The antennas in RU **601** are wirelessly coupled to UE **401** over a 5GNR link. Transceivers in RU **601** are coupled to transceivers in DU **602**. Transceivers in DU **602** are coupled to transceivers in CU **603**. Transceivers in CU **603** are coupled to AMF **422** and UPF **427**. The DSP and CPU in RU **601**, DU **602**, and CU **603** execute operating systems, radio applications, and 5GNR applications to exchange network signaling and user data with UE **401**, AMF **422**, and UPF **427**.

(27) FIG. **7** illustrates exemplary WIFI AN node **412** in 5G wireless communication network **400**. WIFI AN node **412** comprises an example of wireless access nodes **111-113**, although access nodes **111-113** may differ. WIFI AN **412** comprises WIFI radio **701** and node circuitry **702**. WIFI radio **701** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, and transceivers that are coupled over bus circuitry. Node circuitry **702** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in node circuitry **702** stores an operating system and network applications for IP and WIFI. The antennas in WIFI radio **701** are wirelessly coupled to UE **401** over a WIFI link. Transceivers in WIFI radio **701** are coupled to transceivers in node circuitry **702**. Transceivers in node circuitry **702** are coupled to transceivers in IWF **421**. The CPU in node circuitry **702** executes the operating system and network applications to exchange network signaling and user data with UE **401** and with IWF **421**.

(28) FIG. **8** illustrates exemplary data center **420** in 5G wireless communication network **400**. Data center **420** comprises an example of network controller **121**, UDM **122**, UDR **123**, and network element **124**, although controller **121**, UDM **122**, UDR **123**, and element **124** may differ. Data center **420** comprises Network Function (NF) hardware **801**, NF hardware drivers **802**, NF operating systems **803**, NF virtual layer **804**, and NF Software (SW) **805**. NF hardware **801** comprises Network Interface Cards (NICs), CPU, RAM, Flash/Disk Drives (DRIVE), and Data Switches (DSW). NF hardware drivers **802** comprise software that is resident in the NIC, CPU, RAM, DRIVE, and DSW. NF operating systems **803** comprise kernels, modules, and applications that form containers for virtual layer and NF software execution. NF virtual layer **804** comprises

vNIC, vCPU, vRAM, vDRIVE, and vSW. NF SW **805** comprises IWF SW **821**, AMF SW **822**, SMF SW **823**, UDM SW **824**, UDR SW **825**, NEF SW **826**, UPF SW **827**, AUSF SW **828**, HSS SW **829**, CSCF SW **830**, TAS SW **831**, and AF SW **832**. Other NFs like Policy Control Function (PCF) and Network Repository Function (NRF) are typically present but are omitted for clarity. Data center **420** may be located at a single site or be distributed across multiple geographic locations. The NIC in NF hardware **801** are coupled to 5GNR AN **411**, WIFI AN **412**, CDMA network **440**, AS **450**, and external systems. NF hardware **801** executes NF hardware drivers **802**, NF operating systems **803**, NF virtual layer **804**, and NF SW **805** to form and operate IWF **421**, AMF **422**, SMF **423**, UDM **424**, UDR **425**, NEF **426**, UPF **427**, AUSF **428**, HSS **429**, CSCF **430**, TAS **431**, and AF **432**.

(29) FIG. **9** illustrates an exemplary operation of 5G wireless communication network to serve UE **401** and deliver the UE status information for UE **401** upon request. The operation may vary in other examples. UE **401** and AMF **422** exchange registration signaling (REG). The registration signaling reports UE capabilities to AMF **422** like VoP capability, IMS capability, SIP capability, geographic location, and authentication. In response to the registration signaling, AMF **422** and UDM **424** exchange authentication signaling (AUTH) for UE **401**. In response to the authentication of UE **401**, UDM **424** requests (RQ) UE status information for UE **401** from AMF **422**. AMF **422** interacts with UDM **424** to select initial context for UE **401**. AMF **422** interacts with SMF **423** and other network functions to select slices, policies, addresses, and additional context for UE **401**. Part of the context characterizes an IMS bearer between UE **401** and CSCF **430** over UPF **427**. SMF **423** directs UPF **427** to serve the IMS bearer for UE **401** per the context. AMF **422** directs UE **401** to use the IMS bearer over 5GNR AN **411** per the context. UE **401** and CSCF **430** exchange IMS registration signaling over the IMS bearer which traverses UPF **427**. CSCF **430** and HSS **429** exchange IMS registration data for UE **401**. In response to the subscription for UE **401** from UDM **424**, AMF **422** identifies UE status information that indicates: last radio contact, RAT type, VoP capability, IMS capability, SIP capability, geographic location, successful authentication, and/or some other UE metadata. AMF **422** transfers the UE status data for UE **401** to UDM **424** to serve the subscription. UDM **424** transfers the UE status data for UE **401** to UDR **425**. UE **401** places a multimedia call by transferring a SIP invite for the called-party to CSCF **430** over the IMS bearer served by UPF **427**. CSCF **430** transfers the invite to TAS **431**. TAS **431** queries HSS **429** for the UE status information for UE **401** (UE RQ). HSS **429** retrieves the UE status information for UE **401** from UDR **425** and responds to TAS **431** with the UE status information for UE **401**. TAS **431** selects an access network type for UE **401** based on the UE status information for UE **401**. TAS **431** might select a VoNR based on a 5GNR VoP capability for UE **401**. TAS **431** might select CDMA fallback for UE **401** based on an inadequate VoP capability for UE **401**. TAS **431** informs CSCF **430** of the selected access network type for UE **401**. CSCF **430** informs AMF **422** of the selected access network type for UE **401**—typically over a PCF. AMF **422** signals UE **401** to use the selected access network type. UE **401** typically uses the VoNR network that traverses UPF **427** if UE **401** is VoNR capable. UE **401** typically uses CDMA fallback network **440** when UE **401** is not VoNR capable.

(30) FIG. **10** illustrates an exemplary operation of 5G wireless communication network **400** to serve UE **401** and deliver the UE status information for UE **401** upon request. The operation may vary in other examples. UE **401** and AMF **422** exchange registration signaling. The registration signaling reports UE capabilities to AMF **422** like VoP capability, IMS capability, SIP capability, geographic location, and authentication. In response to the registration signaling, AMF **422** and UDM **424** exchange authentication signaling for UE **401**. In response to the authentication of UE **401**, UDM **424** requests UE status information for UE **401** from AMF **422**. AMF **422** interacts with UDM **424** to select initial context for UE **401**. AMF **422** interacts with SMF **423** and other network functions to select slices, policies, addresses, and additional context for UE **401**. Part of the context characterizes an IMS bearer between UE **401** and CSCF **430** over UPF **427**. SMF **423** directs UPF

427 to serve the IMS bearer for UE **401** per the context. AMF **422** directs UE **401** to use the IMS bearer over 5GNR AN **411** per the context. UE **401** and CSCF **430** exchange IMS registration signaling over the IMS bearer which traverses UPF **427**. CSCF **430** and HSS **429** exchange IMS registration data for UE **401**. In response to the subscription for UE **401** from UDM **424**, AMF **422** identifies UE status information that indicates: last radio contact, RAT type, VoP capability, IMS capability, SIP capability, geographic location, successful authentication, and/or some other UE metadata. AMF **422** transfers the UE status data for UE **401** to UDM **424** to serve the subscription. UDM **424** transfers the UE status data for UE **401** to UDR **425**. UE **401** receives a multimedia call when an invite for UE **401** is received by CSCF **430** from another UE or IMS. CSCF **430** and TAS **431** exchange the invite. CSCF **430** forwards the invite to UE **401** over the IMS bearer served by UPF **427**. UE **401** accepts (OK) the invite to CSCF **430** over the IMS bearer that traverses UPF **427**. CSCF **430** transfers the acceptance to TAS **421**. TAS **431** queries HSS **429** for the UE status information for UE **401** (UE RQ). HSS **429** retrieves the UE status information for UE **401** from UDR **425** and responds to TAS **431** with the UE status information for UE **401**. TAS **431** selects an access network type for UE **401** based on the UE status information for UE **401**. TAS **431** might select VoNR based on a 5GNR VoP capability for UE **401**. TAS **431** might select CDMA fallback for UE **401** based on an inadequate VoP capability for UE **401**. TAS **431** informs CSCF **430** of the selected access network type for UE **401**. CSCF **430** informs AMF **422** of the selected access network type for UE **401**—typically over a PCF. AMF **422** signals UE **401** to use the selected access network type. UE **401** typically uses the VoNR network that traverses UPF **427** if UE **401** is VoNR capable. UE **401** typically uses CDMA fallback network **440** when UE **401** is not VoNR capable.

(31) FIG. **11** illustrates an exemplary operation of 5G wireless communication network **400** to serve UE **401** and deliver the UE status information for UE **401** upon request. The operation may vary in other examples. UE **401** and AMF **422** exchange registration signaling. The registration signaling reports UE capabilities to AMF **422** like VoP capability, IMS capability, SIP capability, and geographic location. In response to the registration signaling, AMF **422** and UDM **424** exchange authentication signaling for UE **401**. In response to the authentication of UE **401**, UDM **424** requests UE status information for UE **401** from AMF **422**. AMF **422** interacts with UDM **424** to select initial context for UE **401**. AMF **422** interacts with SMF **423** and other network functions to select slices, policies, addresses, and additional context for UE **401**. Part of the context characterizes a data bearer for UE **401** over UPF **427**. SMF **423** directs UPF **427** to serve the data bearer for UE **401** per the context. AMF **422** directs UE **401** to use the data bearer per the context. UE **401** and external systems exchange user data over the data bearer which traverses UPF **427**. For example, UE **401** may comprise an environmental sensor that is sending carbon dioxide readings over the internet to a central server. In response to the subscription for UE **401** from UDM **424**, AMF **422** identifies UE status information that indicates: last radio contact, RAT type, VoP capability, IMS capability, SIP capability, geographic location, successful authentication, and/or some other UE metadata. AMF **422** transfers the UE status data for UE **401** to UDM **424** to serve the subscription. UDM **424** transfers the UE status data for UE **401** to UDR **425**. AS **450** has some affiliation with UE **401** like employer/employee, parent/child, sensor/owner, or the like. AS **450** logs-in to AF **432** and requests UE status information for UE **401**. AF **432** requests the UE status information for UE **401** from NEF **431**. NEF **431** requests the UE status information for UE **401** from UDR **425**. UDR **425** has the UE status information for UE **401** as described above. UDR **425** transfers the UE status information for UE **401** to NEF **431**. NEF **431** transfers the UE status information for UE **401** to AF **432**. AF **432** transfers the UE status information for UE **401** to AS **450** for use by the employer, parent, owner, or other affiliated entity. For example, the owner of the environmental sensor and central server may want to know where UE **401** is, its RAT type, and when it was last authenticated. UE **401** exchanges additional user data over the data bearer and changes status like a new authentication over a new RAT type at a new geographic location. AMF

422 transfers the new UE status data for UE **401** to UDM **424** to serve the subscription. UDM **424** transfers the new UE status data for UE **401** to UDR **425**. UDR **425** transfers the new UE status information for UE **401** to NEF **431**. NEF **431** transfers the new UE status information for UE **401** to AF **432**. AF **432** transfers the new UE status information for UE **401** to AS **450**.

(32) The wireless data network circuitry described above comprises computer hardware and software that form special-purpose networking circuitry to serve UEs and deliver individual UE status information upon request. The computer hardware comprises processing circuitry like CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory. To form these computer hardware structures, semiconductors like silicon or germanium are positively and negatively doped to form transistors. The doping comprises ions like boron or phosphorus that are embedded within the semiconductor material. The transistors and other electronic structures like capacitors and resistors are arranged and metallically connected within the semiconductor to form devices like logic circuitry and storage registers. The logic circuitry and storage registers are arranged to form larger structures like control units, logic units, and Random-Access Memory (RAM). In turn, the control units, logic units, and RAM are metallically connected to form CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory.

(33) In the computer hardware, the control units drive data between the RAM and the logic units, and the logic units operate on the data. The control units also drive interactions with external memory like flash drives, disk drives, and the like. The computer hardware executes machine-level software to control and move data by driving machine-level inputs like voltages and currents to the control units, logic units, and RAM. The machine-level software is typically compiled from higher-level software programs. The higher-level software programs comprise operating systems, utilities, user applications, and the like. Both the higher-level software programs and their compiled machine-level software are stored in memory and retrieved for compilation and execution. On power-up, the computer hardware automatically executes physically-embedded machine-level software that drives the compilation and execution of the other computer software components which then assert control. Due to this automated execution, the presence of the higher-level software in memory physically changes the structure of the computer hardware machines into special-purpose networking circuitry to serve UEs and deliver individual UE status information upon request.

(34) The above description and associated figures teach the best mode of the invention. The following claims specify the scope of the invention. Note that some aspects of the best mode may not fall within the scope of the invention as specified by the claims. Those skilled in the art will appreciate that the features described above can be combined in various ways to form multiple variations of the invention. Thus, the invention is not limited to the specific embodiments described above, but only by the following claims and their equivalents.

Claims

1. A method comprising: authenticating, by an Access and Mobility Management Function (AMF) in a wireless communication network, a wireless communication device; in response, determining, by the AMF, status information for the wireless communication device; loading, by the AMF, the status information to a Unified Data Registry (UDR) via a Unified Data Management (UDM); receiving, by a Call State Control Function (CSCF) of an Internet Protocol Multimedia Subsystem (IMS), a request for a multimedia call for the wireless communication device and retrieving, by the CSCF, the status information from the UDR via a Home Subscriber System (HSS); selecting, by a Telephony Application Server (TAS) of the IMS, an access network type to support the multimedia call for the wireless communication device based on the status information, wherein the access network type comprises at least one of Fifth Generation New Radio (5GNR) or WiFi; and signaling, by the AMF, the wireless communication device to use the selected access network type

wherein the wireless communication device uses the selected access network type to participate in the multimedia call.

2. The method of claim 1 wherein the status information indicates a time of last radio contact with the wireless communication device.

3. The method of claim 1 wherein the status information indicates a Radio Access Technology (RAT) type serving the wireless communication device.

4. The method of claim 1 wherein the status information indicates a Voice over Packet (VoP) capability for the wireless communication device.

5. The method of claim 1 wherein the status information indicates an IMS capability for the wireless communication device.

6. The method of claim 1 wherein the status information indicates a Session Initiation Protocol (SIP) capability for the wireless communication device.

7. The method of claim 1 wherein the status information indicates a geographic location for the wireless communication device.

8. The method of claim 1 further comprising delivering, by a user plane in the wireless communication network, the multimedia call to the wireless communication device over the selected access network type.

9. A method comprising: exchanging, by an Access and Mobility Management Function (AMF) of a wireless communication network, authentication signaling with a Unified Data Management (UDM) to authenticate a wireless communication device; in response to authentication, determining, by the AMF, status information for the wireless communication device; transferring, by the AMF, the status information to the UDM; loading, by the UDM, the status information on a Unified Data Registry (UDR); receiving, by a Call State Control Function (CSCF) of an Internet Protocol Multimedia Subsystem (IMS), an invite for a multimedia call for the wireless communication device, and requesting, by the CSCF, the status information from a Home Subscriber System (HSS), wherein the wireless communication device generates and transfers the invite for the multimedia call; retrieving, by the HSS, the status information from the UDR and transferring, by the HSS, the status information for delivery to a Telephony Application Server (TAS) of the IMS in response to the request from the CSCF; receiving, by the TAS, the status information from the HSS, and in response, selecting, by the TAS, a Radio Access Technology (RAT) type, wherein the RAT type comprises at least one of Fifth Generation New Radio (5GNR) or WiFi; indicating, by the CSCF, the selected RAT type to the AMF; and signaling, by the AMF, the wireless communication device to use the selected RAT type wherein the wireless communication device uses the selected RAT type to participate in the multimedia call.

10. The method of claim 9 wherein the selected RAT type comprises 5GNR.

11. The method of claim 9 wherein the selected RAT type comprises WiFi.

12. The method of claim 9 wherein selecting, by the TAS, the RAT type comprises selecting, by the TAS, a Voice over New Radio (VoNR) network based on the status information.

13. The method of claim 9 wherein selecting, by the TAS, the RAT type comprises selecting, by the TAS, a Call Division Multiple Access (CDMA) network based on the status information.

14. The method of claim 9 further comprising delivering, by a user plane in the wireless communication network, the multimedia call to the wireless communication device over the selected RAT type.

15. The method of claim 9 wherein the status information indicates one or more of a time of last radio contact with the wireless communication device, a current RAT type serving the wireless communication device, a Voice over Packet (VoP) capability for the wireless communication device, an IMS capability for the wireless communication device, a Session Initiation Protocol (SIP) capability for the wireless communication device, or a geographic location for the wireless communication device.

16. The method of claim 14 wherein the user plane comprises a User Plane Function (UPF).

17. A data communication system comprising: an Access and Mobility Management Function (AMF); a Unified Data Management (UDM); a Unified Data Registry (UDR); a Home Subscriber System (HSS); a Call State Control Function (CSCF) of an Internet Protocol Multimedia Subsystem (IMS); a Telephony Application Server (TAS) of the IMS; and a wireless communication device; the AMF configured to: authenticate the wireless communication device; in response, determine status information for the wireless communication device; and load the status information to the UDR via the UDM; the CSCF configured to: receive a request for a multimedia call for the wireless communication device; and in response, retrieve the status information for the wireless communication device from the UDR via the HSS; the TAS configured to: select a Radio Access Technology (RAT) type for the wireless communication device based on the status information, wherein the selected RAT type comprises at least one of Fifth Generation New Radio (5GNR) or WiFi; and the AMF further configured to: signal the wireless communication device to use the selected RAT type wherein the wireless communication device uses the selected RAT type to participate in the multimedia call.

18. The data communication system of claim 17 wherein the TAS is further configured to select at least one of a Voice over New Radio (VoNR) network or a Call Division Multiple Access (CDMA) network based on the status information.

19. The data communication system of claim 17 further comprising a user plane; and wherein: the user plane is configured to establish a communication link for the multimedia call of the wireless communication device over the RAT type.

20. The data communication system of claim 17 wherein the status information indicates a time of last radio contact with the wireless communication device, a current RAT type serving the wireless communication device, a Voice Over Packet (VoP) capability for the wireless communication device, an IMS capability for the wireless communication device, a Session Initiation Protocol (SIP) capability for the wireless communication device, and a geographic location for the wireless communication device.